

Hybrid Continuum-Tentacle Robotic Manipulator

Project Progress and System Readiness

Veeresh S K, Rishith Gupta, Arshit Singh
Prithvi Sangam Chakravarthy, Arijit Paul

TUE-16
S-08

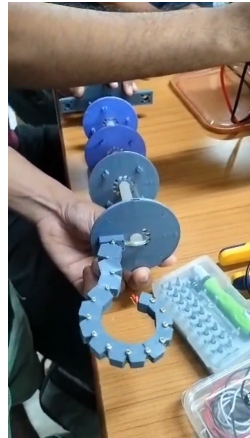
March 21, 2026

Feedback & Actions from Milestone 2

- **Feedback 1:** High friction between nylon line and vinyl tubing
- **Action taken:** Updated CAD design to avoid sharp degree turns
- **Feedback 2:** Mechanical interference in motor-base plate assembly
- **Action taken:** Modified base plate geometry to accommodate new tubing radii
- **Feedback 3:** High slide count and marketing-style presentation
- **Action taken:** Consolidated slides to improve content density, add visual cues and prioritised technical details over decorative formatting

Proof of Concept Demonstration

- Demonstrates working proof of concept
- Shows actuation and basic functionality
- Confirms feasibility of design



Click on the thumbnail for video

PCB Readiness

Current Status

- PCB schematics and layout completed
- Design reviewed and iterated
- Final design submitted to IITB PCB Lab

Key Design Improvements

- Reduced trace lengths for integrity
- Ground planes on both PCB layers
- Improved decoupling placement

Key Learnings

- Proper grounding reduces noise
- Routing impacts signal stability

Step No.	Checklist	Time (hr)	Dimensions (mm)	Dimensions (mm)
1	Initial design review (IITB PCB Lab)	10	100	100
2	Schematic capture (IITB PCB Lab)	10	100	100
3	PCB layout (IITB PCB Lab)	10	100	100
4	PCB fabrication (IITB PCB Lab)	10	100	100
5	PCB assembly (IITB PCB Lab)	10	100	100
6	PCB testing (IITB PCB Lab)	10	100	100
7	PCB documentation (IITB PCB Lab)	10	100	100
8	PCB delivery (IITB PCB Lab)	10	100	100
9	PCB installation (IITB PCB Lab)	10	100	100
10	PCB operation (IITB PCB Lab)	10	100	100
11	PCB maintenance (IITB PCB Lab)	10	100	100
12	PCB disposal (IITB PCB Lab)	10	100	100

Final PCB Layout

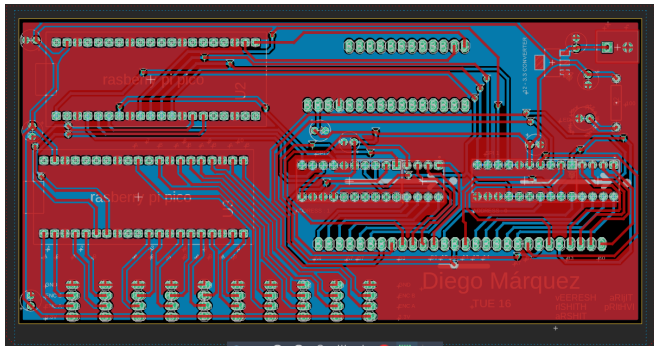


Figure: Final PCB layout with optimized routing and grounding

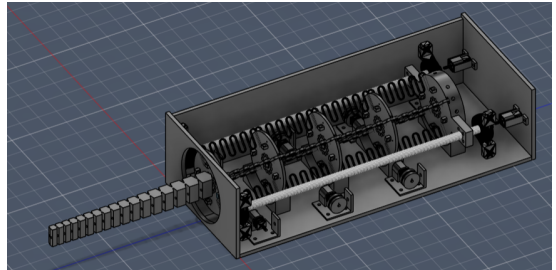
CAD Readiness for Assembly of Prototype

Current Status

- All components fabricated and available
- Test-fitted for clearances and tolerances
- Tendon routing and actuator mounts implemented

Assembly Readiness

- Mounting interfaces finalized
- Cable routing paths defined
- Modular subsystem assembly enabled



Critical Tasks In Progress

- **System Integration:** Electronics routing and subsystem interfacing
- **Control Tuning:** Interrupt-based control with encoder feedback
- **Tendon Calibration:** Tension tuning for symmetric bending
- **System Testing:** End-to-end validation with GUI and camera

Completed Preparatory Work

- Mechanical subsystems tested and partially assembled
- Motors, encoders, and port expanders tested individually
- PCB design completed and sent for fabrication

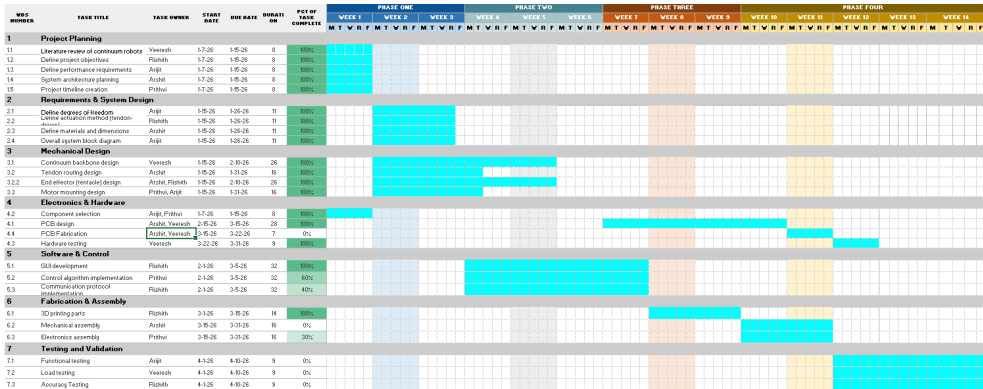
Key Risks Yet To Be Addressed

- **Mechanical Friction:** Joint friction in tentacle segments may hinder smooth motion
- **Tendon Actuation Limits:** Bending paths may reduce effective tendon force
- **Algorithmic Latency:** Control logic may exceed ISR timing limits
- **Encoder Reliability:** Noise or missed counts may affect position accuracy

Mitigation Strategies

- **Incremental Testing:** Validate control loop one axis at a time
- **Timing Profiling:** Verify ISR execution using logic analyzer
- **Mechanical Tuning:** Optimize tendon routing and reduce joint friction

Progress vs Test Plan (Gantt Chart)



- Most milestones on schedule
- Minor delay in Control Algorithm
- Mechanical and electrical design progressed faster than planned

Work Distribution Until Final Demo

Member	Responsibility	Key Tasks
Arshit	Control Algorithm	Interrupt logic, motor control
Prithvi	Control Algorithm	Encoder handling, control tuning
Rishith	Mechanical Assembly	Tentacle assembly and calibration
Veeresh	Mechanical Assembly	Continuum assembly and actuation setup
Arijit	Electrical Assembly	Wiring, PCB integration

- Parallel development of mechanical, electrical, and control subsystems
- Staged integration with testing after each subsystem merge
- Final system-level validation before demonstration

Plan for Final Demonstration

Demonstration Sequence

- Fully integrated hybrid continuum–tentacle robotic system
- User control via GUI with live camera feed
- Controlled multi-directional bending using tendon actuation
- Navigation through a constrained path and object grasping

Evaluation Metrics

- Consistent and repeatable continuum bending response
- Successful object grasping and retrieval in test environment
- Reliable integration of mechanical, electrical, and control subsystems